

Vibrations Workshop 2

By: Jake Peyser

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General Overview

In this assignment, multiple systems involving pendulums, springs, and damping devices will be analyzed. In particular, the systems will be linearized (with $\cos \approx 1$ & $\sin \approx \text{id}$), and the respective natural frequencies will be extracted from the derived equations of motion in terms of the provided parameters.

Lagrangian Mechanics Approach

$$\mathcal{L} = K - V \quad \text{s.t.} \quad K = (\text{kinetic energy}) \quad \text{and} \quad V = (\text{potential energy})$$

Provided generalized coordinates $\{q_i\}$ and a non-conservative force $\vec{F}$ with a respective position arm $\vec{r}$, one has the Euler-Lagrange equation(s) as given below for a single body:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \vec{F} \cdot \frac{\partial \vec{r}}{\partial q_i}$$

Problem (a)

In problem (a), a simple pendulum with rod length l and mass m is given. It will be assumed that the pendulum exists within a uniform gravitational field (with strength g). This is all illustrated in figure 1.

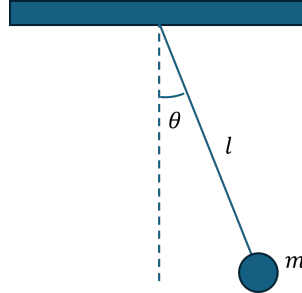


Figure 1: The Simple Pendulum

Setup:

$$q_1 = \theta \quad (\text{which will hold until unstated}) \quad \text{and} \quad K = \frac{1}{2}ml^2\dot{\theta}^2 \quad \text{and} \quad V = -mgl \cos \theta \quad \text{and} \quad \vec{F} = 0$$

The Lagrangian:

$$\mathcal{L} = \frac{1}{2}ml^2\dot{\theta}^2 + mgl \cos \theta$$

Conjugate momentum & force:

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = ml^2\dot{\theta} \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial \theta} = -mgl \sin \theta$$

EOM & natural frequency (ω_n):

$$\frac{d}{dt} [ml^2\dot{\theta}] + mgl \sin \theta = 0 \implies \ddot{\theta} = -\frac{g}{l} \sin \theta \approx -\frac{g}{l} \theta \implies \boxed{\omega_n = \sqrt{\frac{g}{l}}}$$

Observe that ω_n is inversely proportional to $\sqrt{l}$. Also, note that $\frac{\partial \omega_n}{\partial m} = 0$ (i.e. ω_n does not depend on m). This phenomena is due to the equivalence principle, and is a means of detecting local vertical acceleration.

Problem (b)

In problem (b), a disc of mass m and radius r is wound via a coil-like mechanism. That is, rotating the disc from equilibrium ($\theta = 0$) corresponds to a (quadratic) increase in its potential energy. It has a respective (angular) stiffness k as illustrated in figure 2.

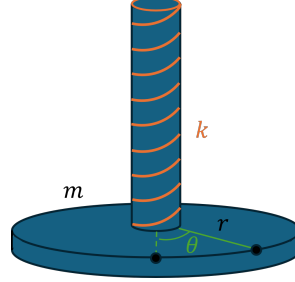


Figure 2: Coiled Rotating Disc

Setup:

$$K = \frac{1}{2}I\dot{\theta}^2 \quad \text{where} \quad I = \frac{1}{2}mr^2 \quad \text{and} \quad V = \frac{1}{2}k\theta^2 \quad \text{and} \quad \vec{F} = 0$$

The Lagrangian:

$$\mathcal{L} = \frac{1}{4}mr^2\dot{\theta}^2 - \frac{1}{2}k\theta^2$$

Conjugate momentum & forces:

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \frac{1}{2}mr^2\dot{\theta} \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial \theta} = -k\theta$$

EOM & anatural frequency (ω):

$$\frac{d}{dt} \left[\frac{1}{2}mr^2\dot{\theta} \right] + k\theta = 0 \implies \ddot{\theta} = -\frac{2k}{mr^2}\theta \implies \boxed{\omega_n = \frac{1}{r}\sqrt{\frac{2k}{m}}}$$

Observe that ω_n is proportional to $\sqrt{k}$, inversely proportional to r , and inversely proportional to $\sqrt{m}$. This radial dependency of ω_n makes the dynamics of this system distinct from a standard mass-spring apparatus.

Problem (c)

Problem (c) depicts the same pendulum as in problem (a) with the same assumptions. The difference comes down to the presence of a horizontal spring attached to the center of the rod. By convention, the spring (with stiffness k) is in equilibrium when the pendulum is vertical.

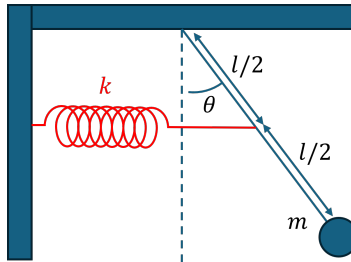


Figure 3: Spring-Enhanced Pendulum

Setup:

$$K = \frac{1}{2}ml^2\dot{\theta}^2 \quad \text{and} \quad V = \frac{1}{2}k\left(\frac{l}{2}\sin\theta\right)^2 - mgl\cos\theta \quad \text{and} \quad \vec{F} = 0$$

Note that the spring is displaced by an amount $\frac{l}{2}\sin\theta$ for any given angular displacement θ , and hence V .

The Lagrangian:

$$\mathcal{L} = \frac{1}{2}ml^2\dot{\theta}^2 - \frac{1}{2}k\left(\frac{l}{2}\sin\theta\right)^2 + mgl\cos\theta$$

Conjugate momentum & force:

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = ml^2\dot{\theta} \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial \theta} = -k\left(\frac{l}{2}\right)^2 \cos\theta \sin\theta - mgl\sin\theta$$

EOM & natural frequency (ω_n):

$$\frac{d}{dt} \left[ml^2\dot{\theta} \right] + k\left(\frac{l}{2}\right)^2 \cos\theta \sin\theta + mgl\sin\theta = 0 \implies \ddot{\theta} = -\frac{k}{4m} \cos\theta \sin\theta - \frac{g}{l} \sin\theta \approx -\left(\frac{k}{4m} + \frac{g}{l}\right)\theta$$

$$\implies \boxed{\omega_n = \sqrt{\frac{k}{4m} + \frac{g}{l}}}$$

Here, the natural frequency takes into account the coupled effects of the spring and the pendulum. That is, setting $k = 0$ transforms the apparatus into a simple pendulum with $\omega_n = \sqrt{\frac{g}{l}}$, and neglecting g transforms the apparatus into a simple mass-spring system with $\omega_n = \frac{1}{2}\sqrt{\frac{k}{m}} \propto \sqrt{\frac{k}{m}}$. Incorporating both components increases the overall ω_n relative to the case of one component in isolation.

Problem (d)

Problem (d) flips problem (c) on its head (literally) and adds the damper component as a means of implementing non-conservative dynamics into the mix. Given the same assumptions as in (c) and the damping coefficient R , the system can be illustrated as in figure 4.

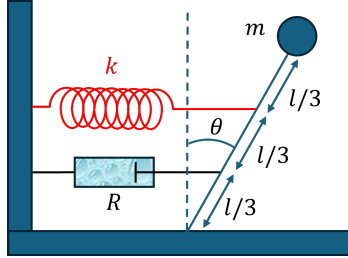


Figure 4: Inverted Pendulum with a Spring-Damping Mechanism

Setup:

$$K = \frac{1}{2}ml^2\dot{\theta}^2 \quad \text{and} \quad V = \frac{1}{2}k\left(\frac{2}{3}l\sin\theta\right)^2 + mgl\cos\theta \quad \text{and} \quad \vec{F} = -R\frac{d}{dt}\left(\frac{l}{3}\sin\theta, 0\right) \quad \text{with} \quad \vec{r} = \frac{l}{3}(\cos\theta, \sin\theta)$$

Note that the spring is displaced by an amount $\frac{2}{3}l\sin\theta$ for any given angular displacement θ , and hence V . Likewise, the damper acts at the point $1/3$ of the way up the rod, and hence $\vec{r}$. Also, the damper only responds to horizontal motion of the pendulum, and hence $\vec{F}$.

The Lagrangian:

$$\mathcal{L} = \frac{1}{2}ml^2\dot{\theta}^2 - \frac{1}{2}k\left(\frac{2}{3}l\sin\theta\right)^2 - mgl\cos\theta$$

Conjugate momentum & force:

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = ml^2\dot{\theta} \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial \theta} = -k\left(\frac{2}{3}l\right)^2 \cos\theta \sin\theta + mgl\sin\theta$$

Generalized (non-conservative) force:

$$\vec{F} \cdot \frac{\partial \vec{r}}{\partial \theta} = \left(-\frac{Rl}{3} \dot{\theta} \cos \theta, 0 \right) \cdot \left(\frac{l}{3} \cos \theta, \frac{l}{3} \sin \theta \right) = -R \left(\frac{l}{3} \right)^2 \dot{\theta} \cos^2 \theta$$

EOM & damped frequency (ω_d):

$$\begin{aligned} \frac{d}{dt} \left[ml^2 \dot{\theta} \right] + k \left(\frac{2}{3} l \right)^2 \cos \theta \sin \theta - mgl \sin \theta &= -R \left(\frac{l}{3} \right)^2 \dot{\theta} \cos^2 \theta \implies \ddot{\theta} + \frac{R}{9m} \dot{\theta} \cos^2 \theta + \frac{4k}{9m} \cos \theta \sin \theta - \frac{g}{l} \sin \theta = 0 \\ \implies \ddot{\theta} + \frac{R}{9m} \dot{\theta} + \left(\frac{4k}{9m} - \frac{g}{l} \right) \theta &\approx 0 \implies \omega_d = \sqrt{\frac{4k}{9m} - \frac{g}{l} - \left(\frac{R}{18m} \right)^2} \end{aligned}$$

By inspection, the notion of ω_d is only valid if $\frac{4k}{9m} > \frac{g}{l} + \left(\frac{R}{18m} \right)^2$. Otherwise, the system is critically damped (=) or overdamped (<). In any case, setting $R = 0$ reveals the natural frequency ω_n of the system:

$$\omega_n = \sqrt{\frac{4k}{9m} - \frac{g}{l}}$$

Just as before, even upon neglecting viscous damping, the (natural) frequency remains unstable. That is to say, if $\frac{g}{l} > \frac{4k}{9m}$, the system does not have a natural frequency with respect to the linearization around $\theta = 0$.

Problem (e)

In problem (e), a set of springs (k_1 & k_2) are placed in series with a mass m in conjunction with a damper (R), which all impact m 's motion in the horizontal direction. By convention, the spring will be in equilibrium when $x = 0$. Gravity (in the vertical direction) will be neglected. This is all depicted in figure 5.

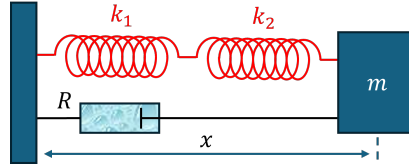


Figure 5: Damper with Springs in Series

In order to write down the system's potential, an effected spring constant k (involving k_1 and k_2) ought to be established first. Because the springs are in series:

$$k = (k_1^{-1} + k_2^{-1})^{-1} = \frac{k_1 k_2}{k_1 + k_2}$$

Setup:

$$q_1 = x \quad \text{and} \quad K = \frac{1}{2} m \dot{x}^2 \quad \text{and} \quad V = \frac{1}{2} k x^2 \quad \text{and} \quad \vec{F} = -R \dot{x} \hat{x} \quad \text{with} \quad \vec{r} = x \hat{x}$$

The Lagrangian:

$$\mathcal{L} = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2$$

Conjugate momentum & force:

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m \dot{x} \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial x} = -kx$$

Generalized (non-conservative) force:

$$\vec{F} \cdot \frac{\partial \vec{r}}{\partial x} = -R \dot{x} \hat{x} \cdot \hat{x} = -R \dot{x}$$

EOM & damped frequency (ω_d):

$$\frac{d}{dt}[m\dot{x}] + kx = -R\dot{x} \implies \ddot{x} + \frac{R}{m}\dot{x} + \frac{k}{m}x = 0 \implies \omega_d = \sqrt{\frac{k}{m} - \left(\frac{R}{2m}\right)^2} = \sqrt{\frac{k_1 k_2}{m(k_1 + k_2)} - \left(\frac{R}{2m}\right)^2}$$

By inspection, the notion of ω_d is only valid if $\frac{k}{m} > \left(\frac{R}{2m}\right)^2$. Otherwise, the system is critically damped ($=$) or overdamped ($<$). In any case, setting $R = 0$ reveals the natural frequency ω_n of the system:

$$\omega_n = \sqrt{\frac{k_1 k_2}{m(k_1 + k_2)}}$$

This is the natural frequency of a double-spring-mass system when in series. Unlike in problem (d), ω_n is always well defined, which indicates the system will oscillate indefinitely/consistently without damping.

Problem (f)

In problem (f), five vertically arranged springs (k_1, k_2, k_3, k_4, k_5) are attached to a mass m as depicted in figure 6. There is uniform gravity (impacting the dynamics in the vertical direction), which will be considered.

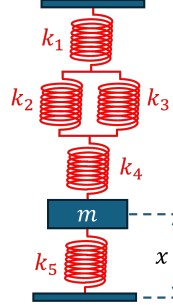


Figure 6: Vertical Springs in Series and Parallel

The springs associated with k_2 and k_3 are in parallel, which could be combined into a single effective spring with stiffness $k_2 + k_3$. Then, the top three (effective) springs are in series. Assuming the net effective spring to be in equilibrium when $x = -mg/k$, the final spring associated with k_5 can be considered to be in parallel with the rest. The net effective stiffness k is thus given as such:

$$k = \left(k_1^{-1} + (k_2 + k_3)^{-1} + k_4^{-1}\right)^{-1} + k_5$$

Setup:

$$q_1 = x \quad \text{and} \quad K = \frac{1}{2}m\dot{x}^2 \quad \text{and} \quad V = \frac{1}{2}k\left(x + \frac{mg}{k}\right)^2 + mgx \quad \text{and} \quad \vec{F} = 0$$

The Lagrangian:

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}k\left(x + \frac{mg}{k}\right)^2 - mgx$$

Conjugate momentum & force:

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x} \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial x} = -kx$$

EOM & natural frequency (ω_n):

$$\frac{d}{dt}[m\dot{x}] + kx = 0 \implies \ddot{x} = -\frac{k}{m}x \implies \omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{\left(k_1^{-1} + (k_2 + k_3)^{-1} + k_4^{-1}\right)^{-1} + k_5}{m}}$$

Overall, this system can be interpreted as a standard spring-mass system with natural frequency ω_n as above.

Conclusionary Remarks

One significant takeaway from this analysis is that many systems involving springs and dampers, regardless of arrangement, can be characterized in terms of the standard spring-mass-damper system. Such systems oscillate/decay predictably. It goes without saying this process is merely a linear approximation of genuine non-linear dynamics, and should be taken as such.

Another takeaway is that alternatives to Newtonian mechanics (i.e. Lagrangian mechanics) is often a valuable tool which can be used to simplify the road to deriving the EOMs of interest. In particular, predominantly working with scalars (i.e. energy) instead of vectors (i.e. forces) eliminates unnecessary geometric analyses.